

Python and Table Readiness

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Classification or Regression?

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

The ML Lifecycle

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Classify records for Rows columns and schemas. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Rows columns and schemas. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Supplied values are 3, 7, 14. Compute their mean. What should a function report for an empty list? How should it handle a missing fourth record?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Trace mean for [4,8,12] using `sum(values)/len(values)`. What should the program do for []? Explain why returning 0 silently is misleading.

6. Solve the given case

Python `lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths)`.

Trace and handle an empty list.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Trace Python: `lengths = [4, 7, 9]; selected = lengths[1]; total = selected + 3`. Give selected and total.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 4 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Create a tiny five-row plant table with explicit column names, types, units, and a reproducible summary calculation.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Trace mean for [4,8,12] using `sum(values)/len(values)`. What should the program do for []? Explain why returning 0 silently is misleading.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 State a schema

List the required fields, types and duration boundaries for the worked table.

My reasoning and evidence

13. W2 Classify records

Classify minutes values 0, 60, -1 and 61 under this schema.

My reasoning and evidence

14. W3 Repair a unit problem

A source says 180 seconds but the row says minutes: 180. Give a documented correction.

My reasoning and evidence

15. W4 Design a schema independently

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

My reasoning and evidence

16. W5 Separate validity and truth

A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?

My reasoning and evidence

17. File input and output — W1 Trace types

Before and after `int("8")`, identify the types and values.

My reasoning and evidence

18. File input and output — W2 Predict the total

A sandbox file has lines 5, 0 and 2. What total does the worked reader print?

My reasoning and evidence

19. File input and output — W3 Repair a path issue

The file is in the class-data folder but the script looks in its working folder. Give a safe repair.

My reasoning and evidence

20. File input and output — W4 Read independently

In a fresh sandbox file counters.txt put 4, 6 and 0 on separate lines. Adapt the reader, print the number of values and total. State the expected results.

My reasoning and evidence

21. File input and output — W5 Handle malformed data

The file contains twelve on one line. Why should the program not silently turn it into zero?

My reasoning and evidence

22. Exceptions and recovery — W1 Trace success

What does `parse_minutes("0")` return, and why is zero allowed?

My reasoning and evidence

23. Exceptions and recovery — W2 Trace two failures

Compare `parse_minutes("abc")` and `parse_minutes("70")`.

My reasoning and evidence

24. Exceptions and recovery — W3 Repair a handler

An except block returns 0 for any failure. State two problems and a repair.

My reasoning and evidence

25. Exceptions and recovery — W4 Build independently

Adapt the function for count strings representing integers 0 through 10. Test "4", "0", "11", "four" and "-1" and record status/value.

My reasoning and evidence

26. Exceptions and recovery — W5 Choose recovery

A batch has 8 valid and 2 invalid lines. What should its report include?

My reasoning and evidence

27. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

28. Pictorial case practice: Python and Table Readiness

Inspect a small table before asking Python to compute an average.

Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/4

Python and Table Readiness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check records

Four rows contain three measured values and one missing value.



2. Calculate mean

Measured sum is $2+6+0=8$; mean is $8/3$ cm, approximately 2.67 cm.



3. Report coverage

Report 3/4 measured, retaining the missing record separately.

CHANGE ONE THING

S changes from 0 to None.

Explain the changed result using the matching lesson.

Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

29. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

30. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Inspect a small table before asking Python to compute an average.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report 3/4 measured, retaining the missing record separately.

- 1. Check records
- 2. Report coverage
- 3. Calculate mean

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Report coverage Report 3/4 measured, retaining the missing record separately. Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: S changes from 0 to None. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Mean becomes $(2+6)/2=4$ cm with $2/4$ measured.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Mean becomes $(2+6)/2=4$ cm with $2/4$ measured. Removing a measured zero changes both the denominator and the coverage.

Repair a missing building block

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review The ML Lifecycle (ai-4/1). Goal: Distinguish fitting a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
